

1. Administrative & Study Identification

Full Study Title: A Global Phase III Study of Rilvegostomig or Pembrolizumab Monotherapy for First-Line Treatment of PD-L1-high Metastatic Non-small Cell Lung Cancer **Short Title/Acronym:** ARTEMIDE-Lung04 **Protocol Number:** D702GC00001 (Secondary IDs: NCI-2025-06305, 2024-517780-24-00) **NCT Number:** NCT06868277 **Trial Status:** RECRUITING **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Rilvegostomig (anti-PD-1/TIGIT bispecific) vs. Pembrolizumab **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of rilvegostomig compared with pembrolizumab monotherapy as a first-line (1L) treatment in participants with metastatic non-small cell lung cancer (mNSCLC) whose tumors express high levels of PD-L1 (Tumor Proportion Score $\geq 50\%$). **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Duration of Response (DoR), and Landmark Survival rates (OS and PFS).

2.2 Background & Rationale Pembrolizumab is a well-established standard of care immunotherapy that targets PD-1 to help the immune system attack cancer. Rilvegostomig is a novel, humanized bispecific IgG1 antibody targeting both PD-1 and TIGIT receptors, which suppress anti-tumor T-cell activity. This study aims to determine whether dual inhibition with rilvegostomig provides superior clinical outcomes (PFS and OS) compared to standard pembrolizumab in patients with PD-L1-high mNSCLC.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab) **Blinding:** Double-blind **Randomization:** 1:1, Randomized, Parallel Assignment

3.2 Planned Enrollment & Duration Target N\$: 830 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 10-Apr-2025 **Estimated Primary Completion:** 17-Jan-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female patients ≥ 18 years of age. 2. Histologically or cytologically documented NSCLC,

including all histological subtypes. Stage IV mNSCLC (based on the American Joint Committee on Cancer Edition 8) not amenable to curative treatment. 3. Tumors express high levels of PD-L1 (confirmed by provision of an acceptable tumor sample for central testing prior to randomization). 4. Absence of sensitizing EGFR (epidermal growth factor) mutations and ALK (anaplastic lymphoma kinase) and ROS1 (c-ros oncogene 1) rearrangements. Negative assay result is required for all non-squamous histology subtypes. 5. Absence of documented tumor genomic mutation results from tests conducted as part of standard local practice in any other actionable driver oncogenes for which there are locally approved targeted 1L therapies. 6. WHO/ECOG performance status of 0 or 1, with no deterioration over the previous 2 weeks prior to baseline. 7. Minimum life expectancy of 12 weeks. 8. At least one lesion not previously irradiated that qualifies as a RECIST 1.1 target lesion at baseline and can be accurately measured at baseline as ≥ 10 mm in the longest diameter (except lymph nodes, which must have short axis ≥ 15 mm) with CT or MRI. 9. Adequate organ and bone marrow function.

4.2 Exclusion Criteria 1. Severe or uncontrolled systemic diseases that make it undesirable for the participant to participate or that would jeopardize compliance with the protocol. 2. Any prior systemic therapy received for advanced or mNSCLC. 3. Prior exposure to anti-PD-1, anti-PD-L1, anti-TIGIT therapy, or any other anticancer therapy targeting immune-regulatory receptors. 4. Active or prior documented autoimmune or inflammatory disorders requiring chronic treatment with steroids or other immunosuppressive treatment. 5. History of organ transplant. 6. History of another primary malignancy except for malignancy treated with curative intent with no known active disease ≥ 2 years before the first dose of study intervention and of low potential risk for recurrence. 7. Presence of small cell and neuroendocrine histology components. 8. Brain metastases unless asymptomatic, stable, and not requiring steroids or anticonvulsants for at least 4 weeks prior to start of study intervention. 9. Active primary immunodeficiency or active infectious disease(s). 10. Active tuberculosis infection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Rilvegostomig (Arm A) or Pembrolizumab (Arm B) administered on a regular dosing cycle. *Pembrolizumab Details:* Trade Name: Keytruda | ATC Code: L01FF02 | EMA URL: [Link](#) **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or completion of maximum treatment duration (up to approximately 5 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease symptoms and adverse events. **Prohibited:** Concurrent treatment with systemic immunosuppressive

therapies (e.g., high-dose corticosteroids), other investigational agents, or live attenuated vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor sample testing for PD-L1, baseline CT/MRI
Baseline	Day 0	Randomization, First Dose
Interim	Every Cycle	Safety Labs, Efficacy Assessment (RECIST 1.1 tumor imaging), PK/PD blood sampling
End of Study	Up to 5 Yrs	Final Vital Signs, Treatment Discontinuation, Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint 1. Overall Survival (OS): Time from randomization until the date of death due to any cause. **2. Progression-Free Survival (PFS):** Time from randomization until radiological progression per RECIST 1.1 or death due to any cause.

7.2 Secondary & Exploratory Endpoints 1. Landmark Overall Survival (OS) rates. 2. Landmark Progression-Free Survival (PFS) rates. 3. Objective Response Rate (ORR). 4. Duration of Response (DoR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of treatment-emergent and immune-related adverse events (irAEs) monitored via investigator assessment and patient reports. **Serious Adverse Events (SAEs):** Standard oncology reporting timelines, typically within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (IDMC) will review safety and efficacy data regularly.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Population for primary efficacy endpoints (PFS and OS). Safety Analysis Set for evaluating AEs. **Sample Size Justification:** $N = 830$, powered to detect a statistically significant superiority in PFS and OS of rilvegostomig over pembrolizumab. **Handling of Missing Data:** Standard survival analysis methodologies, utilizing Kaplan-Meier estimates and censoring rules for progression/survival outcomes.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine on-site and remote monitoring visits to ensure data integrity and protocol adherence. **Audit Trail:** Robust clinical data management including eCRF locking, data clarification forms, and Independent Review Facility (IRF) centralized image collection.

11. Study Identifiers & Governance

Primary ID: D702GC00001 **Registry Number:** NCT06868277 **Posting ID:** 848746526659471250 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARTEMIDE-Lung04 **Official Regulatory Title:** A Phase III, Randomized, Double-blind, Multicenter, Global Study of Riltrogivir or Pembrolizumab Monotherapy for the First-line Treatment of Patients With PD-L1-high Metastatic Non-small Cell Lung Cancer

12. Clinical Framework (Oncology Specific)

Primary Condition: Metastatic Non-small Cell Lung Cancer (mNSCLC) / Carcinoma, Non-Small Cell Lung **Preferred UMLS Name:** PD-L1 Positive Non-Small Cell Lung Carcinoma **Ontology Classifications:** * **Name/Preferred Name:** Carcinoma * **Semantic Type:** Neoplastic Process * **MeSH Code:** D002277 * **HPO Code:** HP: 0030731 * **SNOMED ID:** 1187425009 * **Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas (NCI Thesaurus). **Diagnosis Threshold:** High PD-L1 Expression (Tumor Proportion Score $\geq 50\%$) **Medical Methodology:** First-Line (1L) Immunotherapy **Optimization Target:** Dual targeting of PD-1 and TIGIT receptors

13. Study Procedures & Methodology

Management Pathway: Standard 1L immunotherapy oncology clinical pathway. **Education Protocol (HCP):** RECIST 1.1 training, protocol training, and irAE management protocols. **Education Protocol (Patient):** Informed consent, immunotherapy side-effect monitoring, and disease symptom tracking. **Quality Audit Mechanism:** Centralized imaging review for objective tumor assessments and systematic source data verification.

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 5 years **Onsite Visit Cadence:** Every Treatment Cycle (e.g., Q3W or Q4W dosing schedules depending on specific protocol assignment) **Remote Monitoring Frequency:** Regular survival follow-up contacts after treatment discontinuation **Checklist Review Interval:** Regular tumor assessment scans (typically every 9-12 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]						